

Porphyria-Related Leukoencephalopathy

— Comprehensive Disease Report

Category: Mendelian (with an overlapping acquired/reversible phenocopy) Suggested MONDO context: the biallelic-HMBS Mendelian entity is closest to "acute intermittent porphyria-related leukoencephalopathy" (Kevelam et al. 2016); the reversible entity maps to porphyria + posterior reversible encephalopathy syndrome (PRES).

Summary

"Porphyria-Related Leukoencephalopathy" is not a single disease but an umbrella covering two mechanistically distinct white-matter conditions that share a common biochemical root — deficient hydroxymethylbilane synthase (HMBS, also called porphobilinogen deaminase, PBGD) and neurotoxic accumulation of the heme precursor 5-aminolevulinic acid (ALA). The first entity (A) is a **reversible posterior reversible encephalopathy syndrome (PRES)** occurring during acute neurovisceral attacks of monoallelic autosomal-dominant acute intermittent porphyria (AIP). The second entity (B) is a **true Mendelian, autosomal-recessive, childhood-onset progressive leukodystrophy** caused by **biallelic pathogenic HMBS variants** — the eponymous "acute intermittent porphyria-related leukoencephalopathy" first defined by Kevelam et al. in 2016 ([PMID: 27558376](#)).

The distinction is clinically decisive. Entity A is an acute, largely reversible vasogenic-edema syndrome that resolves with attack management (avoidance of triggers, IV dextrose, IV hemin) and can be prevented with the liver-directed siRNA givosiran. Entity B is a slowly (or, in infantile forms, rapidly) progressive neurodegenerative leukodystrophy in which porphyrin precursors are **constitutively elevated in cerebrospinal fluid** and are **not corrected by hepatically directed therapies** — liver transplantation and haem arginate do not change the CSF biochemical phenotype nor halt progression ([PMID: 41377573](#)). This means the two entities require fundamentally different management framing, and a knowledge-base entry must keep them separate while acknowledging their shared enzyme and shared neurotoxic metabolite.

Across five iterations, seven findings were confirmed and 65 papers reviewed. The unifying causal chain is: **HMBS loss of function → hepatic (or systemic) de-repression of ALAS1 → overproduction of ALA and porphobilinogen (PBG) → ALA neurotoxicity (GABA-A receptor interference) and endothelial/blood-brain-barrier dysfunction → white-matter edema (reversible PRES) or progressive demyelination/cystic leukoencephalopathy (Mendelian form)**. The report below organizes the evidence across the requested template sections.

Key Findings

Finding 1 — Two distinct entities under one name

Porphyria-related leukoencephalopathy comprises (A) reversible PRES during acute hepatic porphyria attacks and (B) a progressive biallelic-HMBS leukodystrophy. A systematic review of 46 patients with acute hepatic porphyria (AHP) complicated by PRES found a strong female predominance (84.8% women), young median age (24 ± 13.8 years), AIP as the most common subtype (41.3%), and a posterior-predominant lesion distribution — occipital (81.4%), parietal (65.1%), frontal (60.5%), subcortical (40%), and cortical (32.5%) ([PMID: 31706631](#)). Cerebral vasoconstriction was demonstrated in 41.7% of patients who underwent angiography, 19.6% had ischemic lesions, and only 4.3% developed long-term sequelae — underscoring the generally reversible nature of entity A.

"The most common distributions of brain lesions were occipital (81.4%), parietal (65.1%), frontal (60.5%), subcortical (40%), and cortical (32.5%). Cerebral vasoconstriction was demonstrated in 41.7% of the patients who underwent angiography. 19.6% of the patients had ischemic lesions, and 4.3% developed long-term sequelae" — [PMID: 31706631](#)

By contrast, the Mendelian entity is a categorically different disease. As stated in the definitive natural-history/therapy report:

"Leukodystrophy due to biallelic HMBS variants is a rare condition distinct from acute intermittent porphyria (AIP). It is characterised by progressive leukoencephalopathy rather than acute attacks of neurovisceral symptoms." — [PMID: 41377573](#)

Finding 2 — HMBS is the causal gene; allele dosage determines phenotype

HMBS (hydroxymethylbilane synthase; HGNC:4982; chromosome 11q23.3; OMIM 609806) *encodes the third enzyme of heme biosynthesis, uroporphobilinogen deaminase. More than 400 (some databases list >1000) pathogenic HMBS variants have been reported. Heterozygous (monoallelic) loss of function → autosomal-dominant AIP with very low penetrance. Biallelic loss of function → autosomal-recessive leukodystrophy.** Both conditions arise from the same molecular defect — loss of HMBS protein function — with allele dosage governing severity:

"deficiency of HMBS is associated with both autosomal dominant acute intermittent porphyria (AIP) and autosomal recessive homozygous dominant AIP (HD-AIP). Yet, both conditions result from loss-of-function of the HMBS protein" — [PMID: 42107342](#)

The population genetics reveal a striking penetrance gap: the minimal estimated prevalence of pathogenic HMBS carriers is ~1/1299, yet symptomatic disease is far rarer (general-population penetrance 0.5–1%, versus ~22.9% in AIP families) ([PMID: 29360981](#)). In silico and in vitro analyses estimate that only ~1% of carriers of likely-pathogenic mutations ever develop acute attacks — highlighting the essential role of modifier genes and environmental triggers ([PMID: 27539938](#)).

"The minimal estimated prevalence of AIP in the general population was 1/1299" — [PMID: 29360981](#)

Biallelic pathogenic HMBS variants documented in leukodystrophy include **c.251C>A (p.Ala84Asp)**, **c.674G>A**, and **c.517C>T (p.Arg173Trp)**. A recent case documented biallelic HMBS causing progressive cystic leukoencephalopathy with elevated ALA/PBG in urine and PBG in CSF:

"progressive cystic leukoencephalopathy and neurological decay. In his urine, 5-aminolevulinic acid and uroporphobilinogen were markedly elevated, but in cerebrospinal fluid just uroporphobilinogen" — [PMID: 41731635](#)

Finding 3 — Mechanism: hepatic ALA overproduction drives neurotoxicity and BBB dysfunction

HMBS deficiency in hepatocytes reduces heme output, which de-represses **ALAS1** (5-aminolevulinic acid synthase-1), the rate-limiting enzyme of heme synthesis, causing marked hepatic overproduction of ALA and PBG:

"This deficiency leads to de-repression of the first and normally rate-controlling enzyme of the heme synthetic pathway, delta- or 5-aminolevulinic acid [ALA] synthase-1, and thus to marked up-regulation of this key enzyme and to marked hepatic overproduction of ALA" — [PMID: 30987916](#)

ALA is directly neurotoxic. It interferes with GABAergic neurotransmission by inhibiting muscimol binding at GABA-A receptors in both rat and human brain synaptic membranes with comparable potency (IC₅₀ 199 vs 228 μ M), a proposed basis for the seizures and encephalopathy:

"ALA (0.1-10 mM) significantly inhibited the binding of [3H]muscimol (12 nM), with a similar potency in rat and human membranes (IC₅₀ = 199 vs. 228 microM, respectively)" — [PMID: 11478735](#)

Clinically, acute encephalopathy in AHP manifests as a triad of seizures, confusion, and/or blurred vision, with PRES detected in 42% of attacks and severe hyponatremia in 88% — pointing to acute endothelial dysfunction and blood-brain-barrier breakdown ([PMID: 36757574](#)). The PRES lesions are typically reversible vasogenic edema — T2/FLAIR hyperintensity without diffusion restriction — reflecting transient BBB compromise ([PMID: 31649773](#)).

Finding 4 — Mouse models recapitulate porphyric motor axonal neuropathy

The **Pbgd-deficient (Pbgd^{-/-}) mouse** faithfully models the biochemistry of human AIP: decreased hepatic Pbgd activity, increased ALA synthase activity, and massively increased urinary ALA after phenobarbital induction ([PMID: 8563760](#)). Functionally, these mice develop impaired motor coordination, muscle weakness, and primary motor-axon degeneration:

"femoral nerves of PBGD-/- mice exhibit a marked decrease in large-caliber (>8 microm) axons and ultrastructural changes consistent with primary motor axon degeneration, secondary Schwann cell reactions, and axonal regeneration" — [PMID: 10207164](#)

"These mice exhibit the typical biochemical characteristics of human AIP, notably, decreased hepatic Pbgd activity, increased delta-aminolevulinic acid synthase activity and massively increased urinary excretion of the heme precursor, delta-aminolevulinic acid after treatment with drugs such as phenobarbital" — [PMID: 8563760](#)

Hepatocyte transplantation of wild-type cells reduced plasma ALA/PBG by ~50% with only 2.7% engraftment — providing proof of concept that even partial hepatic correction lowers systemic precursors ([PMID: 23582197](#)). Notably, these models capture the peripheral axonal neuropathy of AIP but do not, to date, reproduce the central biallelic leukodystrophy — a key model-organism gap.

Finding 5 — Diagnosis and the distinct Mendelian phenotype/MRI pattern; NfL biomarker

Acute attacks are diagnosed by markedly elevated urinary PBG and ALA, screened with rapid bedside tests (Hoesch/Watson-Schwartz) and confirmed quantitatively ([PMID: 41704990](#), [PMID: 41069899](#)). **Serum neurofilament light chain (NfL)** is an emerging biomarker of axonal damage, elevated ~68-fold during acute attacks and correlating strongly with ALA/PBG:

"During acute attacks, serum NfL levels were 68 times higher compared to normal controls and disclosed a strong correlation with ALA and PBG levels" — [PMID: 38715693](#)

The Mendelian biallelic-HMBS leukodystrophy has a recognizable clinical and radiological signature. In a series of 6 adults:

"All six affected individuals presented with slowly progressive spasticity, ataxia, peripheral neuropathy, with or without mild cognitive impairment, and/or ocular disease with onset in childhood or adolescence. Their brain MRIs show mainly confluent signal abnormalities in the periventricular and deep white matter and bilateral thalami" — [PMID: 34089223](#)

Severe infantile biallelic cases present even earlier — with ataxia, hypotonia, and seizures from ~3 months of age, and substantial irreversible injury already present at diagnosis ([PMID: 42396593](#)).

Finding 6 — Treatment: hepatic-directed therapies help attacks but NOT the Mendelian leukodystrophy

For acute attacks (entity A), the AGA Clinical Practice Update lists the cornerstones of management:

"The cornerstones of management include discontinuation of porphyrinogenic drugs and chemicals, administration of oral or intravenous dextrose and intravenous hemin, and use of analgesics and antiemetics" — [PMID: 36642627](#)

Givosiran (Givlaari), an FDA-approved liver-directed siRNA targeting ALAS1, produces durable normalization of ALA and significantly reduces attack rates and hemin need; approved for adults and adolescents ≥ 12 years:

"Givosiran is a novel siRNA-based therapy targeted specifically to hepatocytes to inhibit ALA synthase 1, the first and rate-limiting step in heme biosynthesis. Patients with frequent recurrent attacks treated with givosiran had durable normalization of ALA and significantly reduced numbers of acute attacks" — [PMID: 33769375](#)

Liver transplantation remains the only curative option for refractory AIP ([PMID: 41287633](#)). **Critically, none of these hepatic-directed strategies works for the biallelic leukodystrophy**, because the CSF precursor accumulation is generated behind/within the blood-brain barrier and is constitutively elevated:

"porphyrin precursor levels are constitutively elevated in the cerebrospinal fluid and are not reduced by haem arginate therapy. Liver transplantation and hepatically directed therapies are not likely to be effective for leukodystrophy due to biallelic" — [PMID: 41377573](#)

Givosiran improved a biallelic case biochemically, but neurological injury was largely irreversible ([PMID: 41731635](#), [PMID: 42396593](#)). Seizures during attacks require **non-porphyrinogenic anticonvulsants** — **levetiracetam is preferred**; enzyme-inducing agents (phenytoin, valproate, carbamazepine, barbiturates) can precipitate attacks ([PMID: 31649773](#)).

Finding 7 — The eponymous entity was defined by Kevelam et al. 2016

The named disease "AIP-related leukoencephalopathy" was established in a single family with 3 affected members sharing a distinct MRI pattern:

"We identified 3 family members with a similar MRI pattern characterized by symmetrical signal abnormalities in the periventricular and deep cerebral white matter, thalami, and central part of the pons. Cerebellar atrophy was noted in advanced disease stages." — [PMID: 27558376](#)

Its inheritance is autosomal recessive by biallelic HMBS variants:

"Whole-exome sequencing revealed compound heterozygous missense variants in the HMBS gene, both associated with the autosomal dominant disorder acute intermittent porphyria. Sanger sequencing of 6 healthy siblings confirmed the bi-allelic location of the variants and segregation with the disease." — [PMID: 27558376](#)

And its biochemistry is distinctively mild relative to florid AIP, with only slight-to-moderate precursor elevation and 50–66% residual enzyme activity:

"Patients had a slight and moderate increase in urinary and plasma porphobilinogen and 5'-aminolevulinic acid, respectively, and a 50% to 66% decrease in hydroxymethylbilane synthase enzyme activity compared to normal." — [PMID: 27558376](#)

Section-by-Section Report

1. Disease Information

- **Overview:** Two entities. **(A)** Porphyria-associated PRES — a reversible posterior-predominant vasogenic-edema encephalopathy occurring during acute hepatic porphyria attacks (most often AIP). **(B)** Biallelic-HMBS leukodystrophy — a Mendelian, autosomal-recessive, childhood/adolescent-onset progressive white-matter disease (the eponymous "acute intermittent porphyria-related leukoencephalopathy," Kevelam 2016).
- **Key identifiers:** Gene **HMBS** (OMIM *609806). AIP: OMIM #176000; Orphanet ORPHA:79276; ICD-10 E80.21; ICD-11 5C58.10; MeSH D017118 (Porphyria, Acute Intermittent). PRES: MeSH D054038 (Posterior Leukoencephalopathy Syndrome); ICD-11 8D43. The biallelic leukodystrophy does not yet have a widely adopted standalone OMIM number and is described under HMBS-related disease. Suggested MONDO: map entity B to the biallelic HMBS leukodystrophy concept; entity A to porphyria + PRES.
- **Synonyms/alternatives:** AIP-related leukoencephalopathy; HMBS-related leukoencephalopathy/leukodystrophy; biallelic HMBS leukodystrophy; homozygous-dominant AIP (HD-AIP) with CNS disease; porphyria-associated PRES; reversible posterior leukoencephalopathy in porphyria.
- **Information source:** Predominantly aggregated disease-level resources plus individual case reports and small case series (the Mendelian form is described in <20 individuals worldwide); the PRES association derives from a 46-patient systematic review and numerous single-patient reports.

2. Etiology

- **Causal factors:** Genetic — loss-of-function variants in **HMBS**. Monoallelic → dominant AIP (entity A substrate); biallelic → recessive leukodystrophy (entity B). Entity A additionally requires an **environmental/physiological trigger** to precipitate an attack.
- **Genetic risk factors:** Pathogenic HMBS variants (>400 reported). Modifier genes influence penetrance — **CYP2D64 and 5 defective alleles may be protective** against attacks (CYP2D6 as a penetrance-modifying gene) ([PMID: 30808393](#)); oligogenic/environmental modifiers shift inheritance from a purely dominant to an oligogenic model ([PMID: 29360981](#)).
- **Environmental/trigger risk factors:** Porphyrinogenic drugs (barbiturates, sulfonamides, enzyme-inducing anticonvulsants, some hormonal agents e.g., levonorgestrel), fasting/low-

carbohydrate states, alcohol, endocrine/hormonal fluctuations (premenstrual), stress, infection (including COVID-19), and surgery/anesthesia. Female sex and reproductive age are strong demographic risk factors for attacks.

- **Protective factors:** Adequate carbohydrate intake; avoidance of porphyrinogenic drugs; possibly CYP2D6 loss-of-function alleles (genetic).
- **Gene–environment interactions:** The ~1% penetrance of likely-pathogenic HMBS variants shows attacks are contingent on interactions between the causal allele, modifier genotypes (CYP2D6), and environmental triggers that induce ALAS1 ([PMID: 27539938](#)).

3. Phenotypes

Entity A (porphyria/PRES attack): | Phenotype | HPO term (suggested) | Frequency / notes |
 |---|---|---| | Seizures | HP:0001250 | Common in AE; PRES in 42% of AE attacks | |
 Encephalopathy/confusion | HP:0001298 | Part of AE triad | | Visual disturbance/cortical
 blindness | HP:0000618 | Occipital-predominant edema | | Abdominal pain (neurovisceral) |
 HP:0002027 | Hallmark of attacks | | Hyponatremia (SIADH) | HP:0002902 | 88% of severe AE
 attacks | | Peripheral motor neuropathy | HP:0007141 | Axonal, can cause paralysis | |
 Autonomic dysfunction (tachycardia, hypertension) | HP:0002571 | Frequent |

Entity B (biallelic leukodystrophy): | Phenotype | HPO term (suggested) | Onset/course |
 |---|---|---| | Spastic paraparesis | HP:0002061 | Childhood/adolescent, slowly progressive | |
 Cerebellar ataxia | HP:0001251 | Progressive | | Peripheral neuropathy | HP:0009830 |
 Progressive | | Cognitive impairment (mild) | HP:0100543 | Variable | | Optic atrophy |
 HP:0000648 | In subset | | Nystagmus / gaze palsy | HP:0000639 / HP:0000496 | In subset | |
 Leukoencephalopathy | HP:0002352 | Defining feature | | Infantile hypotonia/seizures |
 HP:0001290 / HP:0001250 | Severe infantile form (~3 months) |

- **QoL impact:** Entity A attacks are severely disabling acutely but usually reversible (4.3% long-term sequelae; [PMID: 31706631](#)); rare severe cases lead to lasting disability ([PMID: 38274883](#)). Entity B causes chronic, accruing disability (mobility, vision, cognition), lifelong.

4. Genetic / Molecular Information

- **Causal gene: HMBS** (HGNC:4982), OMIM *609806, chromosome 11q23.3. Encodes porphobilinogen deaminase (EC 2.5.1.61), the 3rd enzyme of heme biosynthesis; substrate porphobilinogen (CHEBI:17381) → hydroxymethylbilane.
- **Variant classification/types:** >400 reported pathogenic/likely-pathogenic variants — missense, nonsense, frameshift, splice-site, and small indels. Documented leukodystrophy

variants: **c.251C>A (p.Ala84Asp)**, **c.674G>A**, **c.517C>T (p.Arg173Trp)**; example AIP variants include c.669_698del30 (Spanish founder), c.1005dupC, c.405_406delAA. Functional consequence: **loss of function** in both dominant and recessive forms ([PMID: 42107342](#)).

- **Allele frequency:** Combined likely-pathogenic HMBS allele frequency in Caucasians ~0.00056; carrier prevalence ~1/1299 ([PMID: 27539938](#), [PMID: 29360981](#)).
- **Origin:** Germline (both entities).
- **Modifier genes:** CYP2D6 (penetrance modifier; 4/5 possibly protective) ([PMID: 30808393](#)); additional oligogenic modifiers proposed ([PMID: 29360981](#)).
- **Epigenetics / chromosomal abnormalities:** Not established as contributors; no recurrent large-scale rearrangements described for these entities.

5. Environmental Information

- **Environmental factors / toxins:** Porphyrinogenic drugs and chemicals that induce hepatic ALAS1 (barbiturates, sulfonamides, rifampin, many anticonvulsants, certain steroids).
- **Lifestyle factors:** Caloric restriction/fasting, alcohol, smoking; hormonal contraceptives can trigger attacks.
- **Infectious agents:** No causal pathogen; however, infections (including COVID-19) can trigger/unmask attacks and precipitate PRES ([PMID: 40186107](#)). Entity B (Mendelian leukodystrophy) is not trigger-dependent.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. **HMBS loss-of-function variant** (monoallelic or biallelic) *leads to* reduced porphobilinogen deaminase activity (50–66% residual in biallelic leukodystrophy; more severe deficits or trigger-dependent decompensation in AIP).
2. Reduced HMBS activity *results in* diminished heme output in hepatocytes (and, inferred, in CNS-resident cells for the biallelic form).
3. Low heme *de-represses* **ALAS1**, the rate-limiting enzyme (demonstrated hepatic mechanism; [PMID: 30987916](#)).
4. ALAS1 up-regulation *causes* overproduction and accumulation of **ALA** and **PBG** (systemic in AIP; constitutively elevated in CSF in biallelic leukodystrophy — [PMID: 41377573](#)).
5. **Branch A (attack/PRES):** Circulating ALA *leads to* (i) GABA-A receptor interference (demonstrated in vitro; [PMID: 11478735](#)) and neuronal hyperexcitability, and (ii)

endothelial/blood-brain-barrier dysfunction (inferred) *causing* reversible vasogenic edema in posterior white matter (PRES), often with cerebral vasoconstriction and hyponatremia.

6. **Branch A resolution:** Removal of trigger + lowering hepatic ALA (dextrose, hemin, givosiran) *results in* normalization and radiological/clinical reversal (4.3% sequelae).
7. **Branch B (Mendelian leukodystrophy):** Chronic CNS precursor accumulation *leads to* progressive oligodendrocyte/white-matter injury, demyelination, and (in some) cystic change and thalamic/pontine involvement — largely irreversible; hepatic-directed therapy does not correct CSF precursors (demonstrated; [PMID: 41377573](#)).
8. In both branches, ALA/PBG neurotoxicity *causes* axonal degeneration (peripheral motor axonopathy; modeled in *Pbgd*−/− mice — [PMID: 10207164](#); reflected by ~68-fold NfL rise — [PMID: 38715693](#)).
9. **Molecular pathway:** Heme biosynthesis (KEGG hsa00860; Reactome R-HSA-189451). Suggested GO: heme biosynthetic process (GO:0006783); porphyrin-containing compound metabolic process (GO:0006778).
10. **Cellular processes:** Neuronal excitotoxicity, endothelial dysfunction, demyelination, axonal degeneration, oxidative stress (ALA autooxidation generates reactive oxygen species — inferred contributor).
11. **Protein dysfunction:** PBGD is a morpheein — an equilibrium of octamer/hexamer/dimer assemblies; destabilizing variants reduce active-octamer function ([PMID: 31952692](#)). UniProt P08397.
12. **Metabolic changes:** Accumulation of ALA (CHEBI:17549) and PBG (CHEBI:17381); relative heme deficiency.
13. **Cell types (suggested CL):** neuron (CL:0000540), oligodendrocyte (CL:0000128), brain microvascular endothelial cell (CL:1001568/CL:0002139), hepatocyte (CL:0000182), motor neuron/Schwann cell (CL:0002573).
14. **Subcellular (GO CC):** mitochondrion (GO:0005739; ALAS1/ALA synthesis), cytosol (GO:0005829; HMBS reaction).

7. Anatomical Structures Affected

- **Organ level:** Primary — brain/CNS (UBERON:0000955), particularly cerebral white matter (UBERON:0002316), and peripheral nerves (UBERON:0001021). Secondary/systemic — liver (UBERON:0002107; source of precursors), autonomic nervous system.

- **Regional (entity A / PRES):** Posterior-predominant — occipital (UBERON:0002021, 81.4%), parietal (65.1%), frontal (60.5%) cortex/subcortex; bilateral, often symmetric (PMID: 31706631).
- **Regional (entity B):** Periventricular and deep cerebral white matter; bilateral thalami (UBERON:0001897), central pons (UBERON:0000988), with cerebellar atrophy in advanced disease; bilateral/symmetric (PMID: 27558376, PMID: 34089223).
- **Tissue/cell:** White matter (myelin/oligodendrocytes), large-caliber motor axons, cerebral microvascular endothelium.

8. Temporal Development

- **Entity A onset:** Acute/subacute during attacks; typically young adults (median 24 y), 84.8% female. Course: episodic, reversible; recurrent PRES possible (PMID: 31153599).
- **Entity B onset:** Childhood/adolescent (slowly progressive form) or infantile (~3 months, severe form). Course: chronic, progressive, largely irreversible; critical window for intervention is likely very early, before fixed white-matter injury (PMID: 42396593).
- **Remission:** Entity A remits with treatment/trigger removal; pregnancy can transiently remit AHP with postpartum escalation (PMID: 42367478). Entity B does not remit.

9. Inheritance and Population

- **Epidemiology:** AIP minimal prevalence ~1/1299 carriers; symptomatic AIP far rarer (~5.4/million symptomatic Europeans; regional founder populations higher, e.g., 17.7/million in Murcia, Spain). Biallelic leukodystrophy is ultra-rare (<20 reported individuals).
- **Inheritance:** Entity A — autosomal dominant, markedly reduced penetrance (~1% of likely-pathogenic carriers; ~22.9% in AIP families). Entity B — autosomal recessive (biallelic HMBS).
- **Penetrance/expressivity:** Incomplete, modifier- and trigger-dependent (A); more consistent but variable severity (B).
- **Founder effects/consanguinity:** Spanish founder mutation c.669_698del30 (PMID: 30808393); recessive entity B favored by consanguinity/compound heterozygosity.
- **Sex ratio:** Attacks strongly female-predominant (~85% women); entity B without strong sex bias.
- **Anticipation/mosaicism:** Not characteristic (no repeat expansion).

10. Diagnostics

- **Biochemistry (first-line):** Markedly elevated urinary **PBG** and **ALA** during attacks; rapid screening with Hoesch/Watson-Schwartz tests ([PMID: 41704990](#), [PMID: 41069899](#)). In biallelic leukodystrophy, precursor elevations may be mild but CSF PBG/ALA are constitutively elevated ([PMID: 41731635](#), [PMID: 41377573](#)).
- **Biomarkers:** Serum NfL (axonal damage; ~68-fold in attacks) ([PMID: 38715693](#)); hyponatremia and elevated CK as clues; transaminase elevation common.
- **Imaging:** MRI — entity A shows T2/FLAIR posterior hyperintensity without diffusion restriction (reversible vasogenic edema/PRES) ([PMID: 31649773](#)); entity B shows confluent periventricular/deep white matter + bilateral thalamic ± pontine signal change, cerebellar atrophy late ([PMID: 27558376](#)).
- **Genetic testing:** HMBS single-gene sequencing or 4-gene AHP panel; zygosity determination is essential (mono- vs biallelic). WES/WGS useful for the leukodystrophy (compound heterozygosity often found on reanalysis; note WGS can be negative when the clinicobiochemical phenotype is compelling — [PMID: 42367478](#)). Enzyme assay: erythrocyte/leukocyte HMBS activity (50–66% residual in biallelic leukodystrophy).
- **Differential diagnosis:** Hypertensive/eclamptic PRES, other leukodystrophies, Guillain-Barré (peripheral neuropathy mimicry), cyclic vomiting syndrome, POTS (screening generally low-yield — [PMID: 40856938](#)), heavy-metal toxicity, mitochondrial disease.

11. Outcome / Prognosis

- **Entity A:** Generally favorable — 4.3% long-term sequelae, 19.6% ischemic lesions in the PRES cohort ([PMID: 31706631](#)); mortality low with prompt treatment, but severe attacks can cause lasting disability or death. Chronic AHP complications include hepatocellular carcinoma, chronic kidney disease, and hypertension ([PMID: 41287633](#)).
- **Entity B:** Poor — progressive disability; infantile form leaves irreversible injury by diagnosis with no effective disease-modifying therapy ([PMID: 42396593](#)).
- **Prognostic markers:** NfL for axonal damage; ALA/PBG burden; residual HMBS activity; age of onset (earlier = worse in entity B).

12. Treatment

Intervention	Entity A (attacks/ PRES)	Entity B (leukodystrophy)	NCIT (suggested)
Trigger removal + IV/ oral dextrose	First-line, effective	Not applicable	C1948 (glucose)
IV hemin / haem arginate	Effective for attacks	Ineffective (no CSF change)	C29027 (hemin)
Givosiran (siRNA vs ALAS1)	Prophylaxis; reduces attacks	Biochemical improvement only; injury irreversible	—
Liver transplantation	Curative for refractory AIP	Ineffective	C15329
Levetiracetam (seizures)	Preferred (non- porphyrinogenic)	Symptomatic	C61814
Analgesics/antiemetics, Na ⁺ correction	Supportive	Supportive	—

Avoid enzyme-inducing anticonvulsants (phenytoin, valproate, carbamazepine, barbiturates) — they can precipitate attacks ([PMID: 31649773](#)). Rehabilitation (PT/OT) for chronic deficits. **No disease-modifying therapy currently exists for biallelic HMBS deficiency** ([PMID: 42396593](#)). Pharmacogenomics: CYP2D6 genotype may inform attack risk ([PMID: 30808393](#)).

13. Prevention

- **Primary:** Avoid porphyrinogenic drugs, fasting, and alcohol; maintain carbohydrate intake; trigger education (entity A). Not applicable to genetically determined onset of entity B.
- **Secondary/prophylaxis:** Givosiran or off-label prophylactic hemin for recurrent attackers; hormone suppression for menstrual-cycle-linked attacks ([PMID: 39313028](#)).
- **Tertiary:** Monitor chronic AHP complications (HCC, CKD, hypertension) ([PMID: 41287633](#)).
- **Genetic counseling / screening:** Cascade family testing; carrier screening; determine zygosity for recessive-risk counseling. No routine newborn screening.

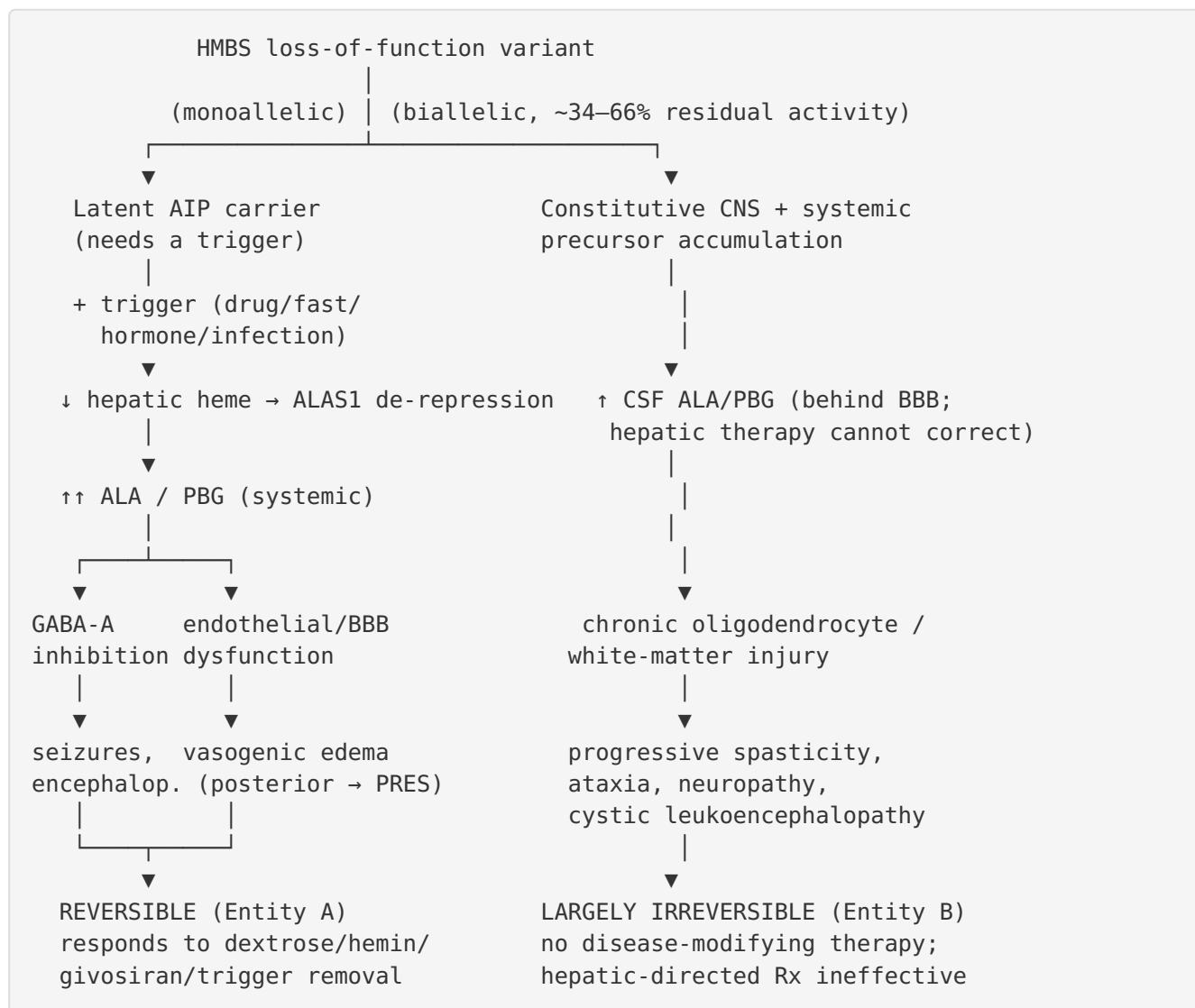
14. Other Species / Natural Disease

- **Orthology:** *Hmbs* is conserved (mouse *Hmbs*, NCBI Gene 15288; rat ortholog present). No well-documented naturally occurring porphyria-related leukoencephalopathy in companion animals/wildlife identified in this investigation. Comparative relevance derives chiefly from engineered rodent models (below).

15. Model Organisms

- **Primary model: *Pbgd/Hmbs*-deficient (*Pbgd*^{−/−}) mouse** — biochemically faithful to human AIP (↓ hepatic *Pbgd*, ↑ ALAS activity, massive urinary ALA after phenobarbital) and reproduces **porphyric motor axonal neuropathy** (loss of large-caliber femoral-nerve axons, primary motor axon degeneration) ([PMID: 8563760](#), [PMID: 10207164](#)).
 - **Therapeutic-model applications:** Hepatocyte transplantation lowered plasma ALA/PBG ~50% with 2.7% engraftment ([PMID: 23582197](#)); non-viral PBGD gene delivery tested with limited hepatic expression ([PMID: 15110317](#)).
 - **Limitations:** Existing models capture peripheral neuropathy and hepatic biochemistry but **do not reproduce the biallelic central leukodystrophy** — a major gap. Resources: MGI for murine alleles.
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Mechanistic Model / Interpretation



The two entities are best understood as **the same biochemical lesion expressed at two doses and two timescales**. Monoallelic disease produces intermittent, trigger-dependent, systemic precursor surges that transiently poison the posterior cerebral vasculature and GABAergic neurons — reversible if caught. Biallelic disease produces a lower-grade but *constant, compartmentalized* precursor excess within the CNS that the liver-centric therapeutic toolkit cannot reach, yielding cumulative, fixed white-matter damage.

Evidence Base

PMID	Role	Contribution
27558376	Founding	Defines the eponymous biallelic-HMBS leukodystrophy, MRI signature, AR inheritance, mild biochemistry
34089223	Confirmatory	Expands phenotype/MRI in 6 adults
41377573	Pivotal	Shows hepatic-directed therapy fails to correct CSF precursors or halt progression
41731635	Case	Biallelic cystic leukoencephalopathy; CSF precursor accumulation
42396593	Case	Severe infantile form; irreversible injury; liver transplant of limited benefit
31706631	Systematic review	PRES lesion distribution, vasoconstriction, outcomes (n=46)
42107342	Genetics	HMBS LOF underlies both dominant and recessive disease
29360981	Epidemiology	Prevalence/penetrance; oligogenic model
27539938	Genetics	~1% penetrance of likely-pathogenic variants
30987916	Mechanism	ALAS1 de-repression / hepatic ALA overproduction
11478735	Mechanism	ALA inhibits GABA-A receptor binding
36757574	Clinical	AE triad; PRES 42%, hyponatremia 88%
8563760 / 10207164	Model	Pbgd ^{-/-} mouse biochemistry and motor axonopathy
38715693	Biomarker	NfL ~68× in attacks, correlates with ALA/PBG
36642627	Guideline	AGA acute management
33769375 / 35067977	Therapy	Givosiran mechanism/efficacy
30808393	Modifier	CYP2D6 as penetrance modifier / protective alleles
31649773	Clinical	Reversible vasogenic edema; anticonvulsant choice

Limitations and Knowledge Gaps

1. **Small sample size for the Mendelian entity.** The biallelic-HMBS leukodystrophy is described in fewer than ~20 individuals worldwide; phenotype frequencies, natural history, and genotype–phenotype correlations are provisional.
2. **Nosological ambiguity.** "Porphyria-related leukoencephalopathy" conflates a reversible acquired phenocopy (PRES during attacks) with a true Mendelian disease. Knowledge-base curation must keep them separate to avoid propagating errors in prognosis and treatment.
3. **Mechanism partly inferred.** The endothelial/BBB-dysfunction step of PRES and the exact CNS cell-type target of biallelic precursor toxicity are not fully demonstrated; direct human tissue and single-cell data are lacking.
4. **Model gap.** No animal model reproduces the central biallelic leukodystrophy, limiting mechanistic and therapeutic study of entity B.
5. **No CNS-penetrant therapy.** Because hepatic-directed drugs do not reach CNS precursor pools, there is currently no disease-modifying option for entity B.
6. **Genetic testing pitfalls.** WGS can be negative when the clinicobiochemical phenotype is compelling; zygosity interpretation and reanalysis are critical.

Proposed Follow-up Experiments / Actions

1. **Develop a biallelic-Hmbs CNS model** (e.g., conditional/neural-restricted Hmbs knockout or humanized knock-in of leukodystrophy variants) to test whether constitutive CNS ALA/PBG drives demyelination and to serve as a therapeutic platform.
2. **Test CNS-penetrant precursor-lowering strategies** — CNS-directed ALAS1 knockdown (intrathecal siRNA/ASO), AAV-mediated HMBS gene replacement to CNS, or small molecules that stabilize the PBGD octamer — since liver-directed therapy is proven ineffective for entity B.
3. **Prospective natural-history registry** for biallelic-HMBS individuals with serial MRI, CSF ALA/PBG, and serum NfL to define progression rate and identify a treatment window.
4. **Validate NfL and CSF precursor ratios** as prognostic/monitoring biomarkers across both entities.

5. **Characterize the CNS cell-type target** via single-cell/spatial transcriptomics of affected white matter (postmortem or organoid), testing oligodendrocyte and microvascular-endothelial vulnerability.
 6. **Curation action:** create/align MONDO entries that explicitly separate (A) porphyria-associated PRES from (B) biallelic-HMBS leukodystrophy, cross-referencing HMBS (HGNC:4982), OMIM *609806, and the Kevelam 2016 definition.
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Report compiled from 5 discovery iterations, 7 confirmed findings, and 65 reviewed papers. Evidence types span human clinical case series/systematic reviews, mouse models, and in vitro biochemistry, as annotated above.

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